

(a) detector constants

quantity	unit	APEX	Python ccdproc	IRAF ccdproc
gain	e-/ADU	0.681 ± 0.014	0.681 ± 0.014	0.681 ± 0.014
read noise	e-	2.35	2.35	2.35
dark current	e-/s	0.0077	0.0077	0.0077

same raw calibration set · same PTC and dark-ladder estimators
 pixel-level residuals are given in Figure 4; no header/vendor value is used

